



KTH Computer Science
and Communication

Exam in DD2421 Machine Learning 2026-03-11, kl 14.00 – 18.00

Aids allowed: *calculator, language dictionary.*

To take this exam you must be registered to this specific exam as well as to the course.

In order to pass this exam, your score x first needs to be 20 or more (out of 42, full point). In addition, given your points y from the Programming Challenge (out of 18, full point), the requirements on the total points, $p = x + y$, are preliminarily set for different grades as:

$$54 < p \leq 60 \rightarrow A$$

$$48 < p \leq 54 \rightarrow B$$

$$42 < p \leq 48 \rightarrow C$$

$$36 < p \leq 42 \rightarrow D$$

$$29 < p \leq 36 \rightarrow E \text{ (A pass is guaranteed with the required points for 'E'.)}$$

$$0 \leq p \leq 29 \rightarrow F$$

This exam consists of sections **A**, **B**, and **C**.

NB: use different papers (answer sheets) for different sections A, B, and C.

This means that you shall start a new page after section A, and again after section B.

A Graded problems

Potential inquiries to be addressed to Atsuto Maki.

A-1 Terminology

(4p)

For each term (a–d) in the left list, find the explanation from the right list which *best* describes how the term is used in machine learning.

- | | |
|--------------------------------|--|
| | 1) The expected amount of information produced by a random variable |
| | 2) A technique that randomly removes input features to reduce dimensionality |
| a) Dropout | 3) An approach to train artificial neural networks |
| b) RANSAC | 4) Algorithm to learn with latent variables |
| c) Occam's razor | 5) A robust method for estimating model parameters in the presence of outliers |
| d) Shannon information content | 6) A measure of the information associated with an event of probability p |
| | 7) A technique that reduces model complexity to prevent overfitting |
| | 8) A principle to choose the simplest explanation |

A-2 Nearest Neighbor, Classification

(5p)

Suppose we divide a dataset into two equal parts, Part I and Part II, and perform 2-fold cross-validation. First we train on Part I and evaluate on Part II. Then we swap the roles of the two parts.

- a) First we use the 1-Nearest Neighbor rule (1-NN) and get an average error rate (averaged over the training and validation errors) of 8%. What was the error rate with 1-nearest neighbor on the validation set? Briefly reason the answer. (1p)
- b) Next we use Decision Trees and get an error rate of 8% on the training data. We also get the average error rate (averaged over the training and validation errors) of 10%. What was the error rate with Decision Trees on the validation set? Briefly reason the answer. (1p)
- c) Now we swap the roles of Part I and Part II and repeat the same experiments. On the validation set (Part I), we get an error rate of 10% with both 1-NN and Decision Trees. Compute the 2-fold cross-validation error for both methods. (2p)
- d) Based on these cross-validation errors, indicate which method should be preferred for classifying new observations and briefly justify your answer. (1p)

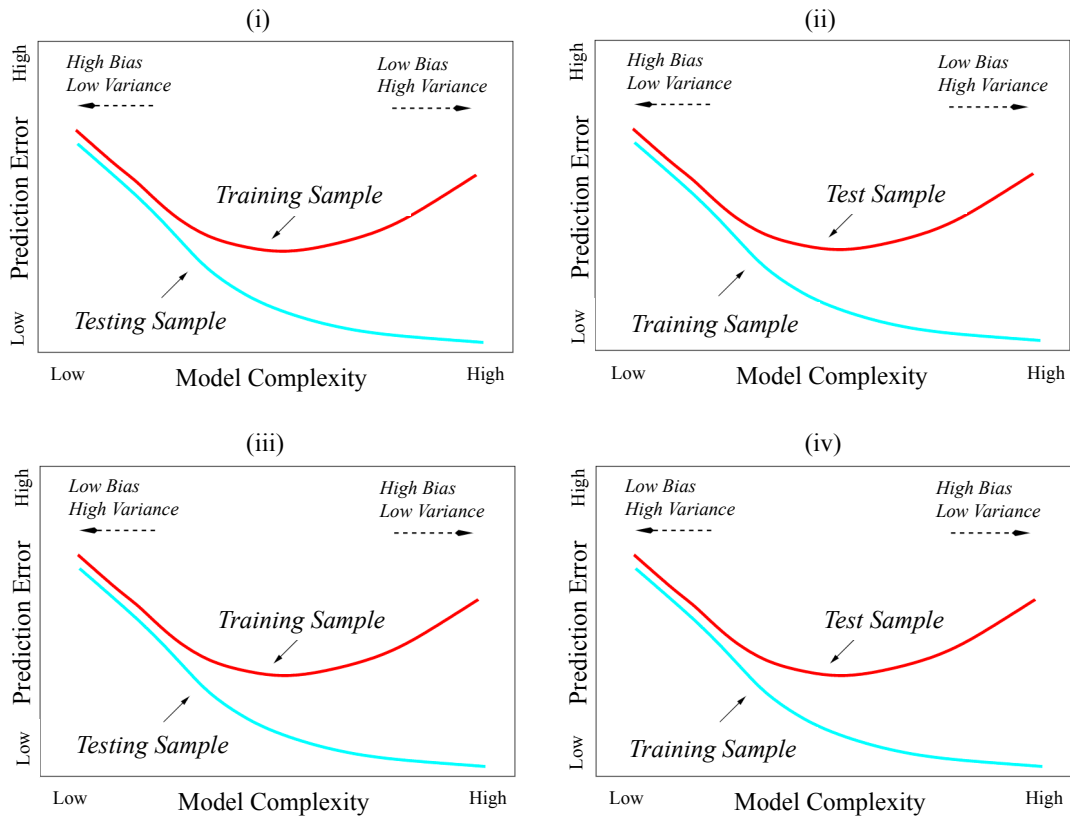


Figure 1. Typical behavior of prediction error plotted against model complexity.

A-3 Bias and Variance trade-off

(5p)

- a) One of the four subfigures (i)-(iv) in Figure 1 displays the typical trend of prediction error of a model for training and testing data with comments on its bias and variance, {high, low}. Which one of the four figures best represents the general situation? (1p)

- b) Now, let us consider a classifier represented by the function $\hat{f}(\mathbf{x})$ of input vector $\mathbf{x}$, and the following concepts:

$\hat{f}(\mathbf{x})$: prediction function (i.e. the model) estimated with a set of data samples, $\mathcal{D}$
 $E_{\mathcal{D}}[\hat{f}(\mathbf{x})]$: the average of models due to different sample sets

Expectations are taken over the sampling of the training set $\mathcal{D}$.

Write expressions for the bias and variance of the estimator $\hat{f}(\mathbf{x})$ in terms of the quantities defined above. (2p)

- c) Derive that the expected squared error $E_{\mathcal{D}}[(\hat{f}(\mathbf{x}) - f(\mathbf{x}))^2]$ can be decomposed into bias and variance. (2p)

A-4 Ensemble Methods

(4p)

Briefly answer the following questions regarding ensemble methods for classification.

- a) What are the two sources of randomness in Random Forests? (2p)
- b) In the AdaBoost algorithm, each training sample is assigned a weight that is updated during iterative training of weak classifiers. What are the two most dominant factors in updating the weights? How are they used? (2p)

A-5 PCA, Subspace Methods

(4p)

We consider solving a K -class classification problem with the Subspace Method. For this purpose we compute a subspace $\mathcal{L}^{(j)}$ ($j = 1, \dots, K$) using training data for each class, respectively. That is, given a set of feature vectors (as training data) which belong to a specific class C (i.e. with an identical class label), we perform PCA on them and select p orthonormal eigenvectors $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ which span a p -dimensional subspace, $\mathcal{L}$, as the outcome. Answer the following questions.

- a) In PCA, eigenvectors corresponding to the largest eigenvalues are selected to form the subspace $\mathcal{L}$. Briefly explain why these eigenvectors provide a good representation of the class data. (2p)
- b) We have a new input vector $\mathbf{x}$ whose class is unknown, and consider the squared length of the projection of $\mathbf{x}$ onto $\mathcal{L}$. Describe how the length of the projection is represented, using a simple formula. (1p)
- c) Given $\mathbf{x}$, we compute its projection length on each subspace as $S^{(j)}$ ($j = 1, \dots, K$). Among those $S^{(\alpha)}$, $S^{(\beta)}$, and $S^{(\gamma)}$ are the smallest, the second smallest, and the largest, respectively. Based on this observation, to which class should $\mathbf{x}$ be assigned? (1p)

B Graded problems

Potential inquiries to be addressed to André Pereira.

B-1 Bayesian Decision Theory: MAP and Conditional Independence

(4p)

You are designing a binary classifier for a security system. Let $C = 1$ denote an *Intruder* and $C = 0$ denote an *Authorized User*. The system outputs an alarm $A = 1$ or silence $A = 0$. You are given:

$$P(C = 1) = 0.01, \quad P(A = 1 | C = 1) = 1, \quad P(A = 1 | C = 0) = \alpha.$$

a) Posterior probability (1p)

Using Bayes' theorem, derive an expression for the posterior probability

$$P(C = 1 | A = 1)$$

as a function of α . Show your steps.

b) MAP decision boundary (2p)

The Maximum A Posteriori (MAP) rule predicts *Intruder* if

$$P(C = 1 | A = 1) > P(C = 0 | A = 1).$$

Derive the exact range of α for which an alarm is classified as an Intruder.

c) Conditional independence (1p)

A second identical sensor is installed. Let A_1 and A_2 denote the outputs of the two sensors. Assume the sensors are conditionally independent given the class C , and that each sensor has the same conditional probabilities as above.

Which of the following best explains why using two sensors allows a higher individual false alarm rate than in part (b)?

- (a) Because the prior probability of an intruder increases.
- (b) Because the probability of two simultaneous false alarms becomes α^2 , which is much smaller than α .
- (c) Because MAP ignores the prior probabilities when two sensors are used.
- (d) Because the sensors always produce identical outputs.

B-2 MAP Estimation and Regularization

(4p)

Consider the probabilistic linear regression model

$$y_i = \mathbf{w}^T \mathbf{x}_i + \epsilon_i$$

for a dataset

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N.$$

Assume the noise ϵ_i follows a Gaussian distribution $\mathcal{N}(0, \sigma^2)$. The likelihood for a single observation is given by:

$$p(y_i | \mathbf{x}_i, \mathbf{w}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{w}^T \mathbf{x}_i)^2}{2\sigma^2}\right).$$

We place an independent Laplace prior on each weight w_d for $d = 1, \dots, D$:

$$p(w_d) = \frac{1}{2b} \exp\left(-\frac{|w_d|}{b}\right).$$

a) Derive the objective function (2p)

Starting from the Gaussian likelihood and the Laplace prior, derive the negative log-posterior.

Show that the MAP estimate w_{MAP} minimizes an objective of the form:

$$\text{SSE} + \lambda \sum_{d=1}^D |w_d|$$

where

$$\text{SSE} = \sum_{i=1}^N (y_i - \mathbf{w}^T \mathbf{x}_i)^2.$$

Express λ in terms of σ^2 and b .

b) Common name (1p)

What is the standard machine learning name for regression with a Laplace prior on the weights?

c) Effect of the prior (1p)

Which of the following best explains why a Laplace prior encourages sparse solutions?

- (a) The Laplace prior increases the variance of the weights.
- (b) The Laplace prior leads to an L_1 penalty with sharp corners at zero, making coefficients more likely to become exactly zero.
- (c) The Laplace prior removes the likelihood term from the optimization.
- (d) The Laplace prior forces all weights to be equal.

B-3 K-Means Clustering and Local Minima

(4p)

Consider the dataset

$$\mathcal{D} = \{1, 2, 8, 9\}.$$

We cluster the data into $K = 2$ clusters using K-Means.

Initial centers:

$$\mu_1 = 1 \quad \mu_2 = 2.$$

a) First iteration (1p)

Assign each data point to the nearest cluster center and compute the updated centers.

b) Second iteration (1p)

Using the updated centers, perform another assignment step and compute the new centers.

c) Third iteration / convergence check (1p)

If you run one more iteration, do any assignments change? State whether K-Means has converged and briefly justify.

d) Convergence and objective (1p)

Which statement is correct?

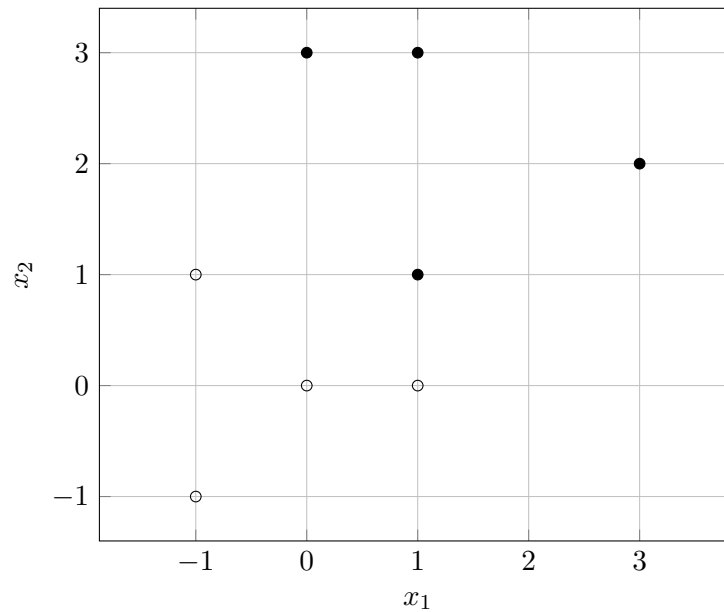
- (a) K-Means is guaranteed to find the global minimum of its objective for any initialization.
- (b) K-Means is guaranteed to converge in one iteration.
- (c) K-Means is guaranteed to converge to a (possibly local) minimum of its objective, so in practice we often use techniques such as multiple random restarts.
- (d) K-Means may increase its objective from one iteration to the next.

C Graded problems

Potential inquiries to be addressed to Jörg Conradt.

C-1 Support Vector Machines

(5p)



The eight datapoints in the figure are used to train a support vector machine. Filled circles are positive samples and unfilled circles are negative samples.

When a linear kernel ($\mathcal{K}(\vec{x}, \vec{y}) = (\vec{x} \cdot \vec{y} + 1)$) was used, and no slack was allowed ($C = \infty$), this resulted in three support vectors, with the corresponding α -values 0.5, 2, and 2.5.

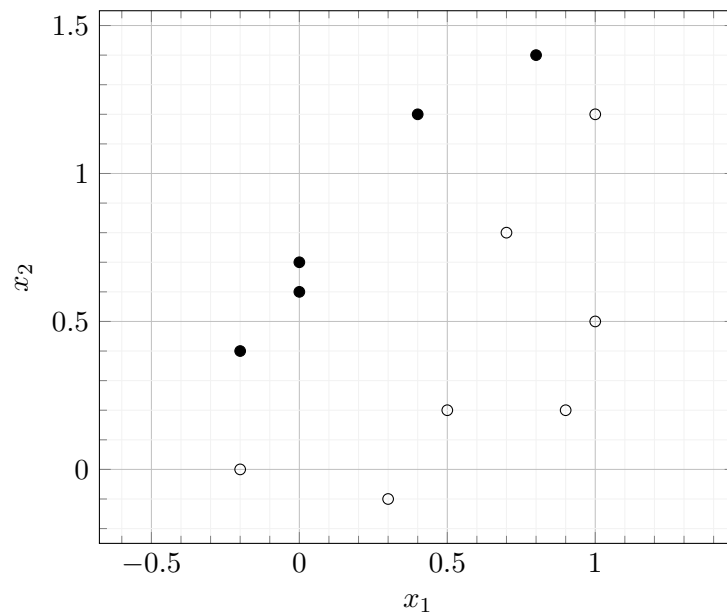
To classify new points, the normal indicator function is used:

$$\text{ind}(\vec{x}) = \sum_k t_k \alpha_k \mathcal{K}(\vec{x}, \vec{s}_k) - b$$

- What will the three support vectors be? You do not need to calculate this mathematically; use your knowledge about support vectors to deduce which three points must be support vectors. (2p)
- Which α -value must be associated with each of the three support vectors? (Give short motivations) (2p)
- What value for b should be used in the indicator function? (1p)

C-2 Linear classifier, single neuron

(3p)



- a) The figure illustrates a dataset where filled circles are positive samples and unfilled are negative. For a single “artificial neuron”, find a set of values for the weights and the threshold that will separate the positive from the negative samples. Make sure that positive points are *above* the threshold and negative points are *below*. (2p)

Note: There is no need to use any learning algorithm here. It is sufficient to find a suitable linear separator in the figure, but you must explicitly state the weight and threshold values and explain how you arrived at these values in keywords or a short text.

- b) If a new negative sample at $(-0.2, 0.5)$ arrives, and learning is done with the *perceptron learning rule*; how will these values change? Do not compute exact values, but explain in keywords or in a short text what happens. (1p)